



CMPS 451

Data Mining, Big Data & Analytics

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Idea

Streaming platforms and media enterprises regularly acquire movies and TV series to improve viewer satisfaction and retention. However, acquisition decisions are often based on limited manual judgment, brand reputation, or past popularity rather than systematic predictive analysis. The business problem addressed in this project is: **How can a streaming enterprise estimate the likely audience reception of a title before acquiring it?**

This project proposes a predictive analytics system that helps acquisition managers and content strategists make better content selection decisions. By analyzing large-scale historical metadata about movies and series, the system will identify the characteristics associated with stronger audience reception and provide data-driven support for prioritizing acquisition candidates.

Datasets

We will use the Official IMDb Non-Commercial Datasets, which provide large-scale real-world data suitable for big data analytics. These datasets contain millions of records describing titles, ratings, genres, runtimes, cast members, directors, and other contributors.

[Link to the dataset](#)

Planned Approach

The project will be implemented using Apache PySpark in pseudo-distributed mode, as required by the project document.

The planned pipeline is:

1. **Data preprocessing:** clean missing and inconsistent values, filter irrelevant records, and join the selected IMDb tables using Spark SQL.
2. **Feature engineering:** generate features such as genre indicators, runtime, title type, release period, and aggregated statistics related to directors and cast members.
3. **Predictive modeling:** train supervised learning models to estimate audience reception, either by predicting the rating directly or classifying titles into rating groups. Candidate models include Multiple Linear Regression, Decision Tree, and Random Forest using Spark MLlib.
4. **Segmentation and insights:** apply K-Means clustering to group titles with similar characteristics and identify patterns associated with highly rated content.
5. **Evaluation and visualization:** measure model performance on training and testing data and present visual insights useful for business decision-making.